Biochar amendment effects on grain yield and Fusarium crown rot severity in winter wheat: a greenhouse study

SYNTHETIC EXAMPLE - all text, data and references are invented to test a peer-review skill. Not a real study.

Abstract

Biochar is widely promoted as a soil amendment that can improve crop yield and suppress soil-borne disease. We tested four biochar rates (0, 5, 10 and 20 t ha-1) in a greenhouse pot experiment with winter wheat inoculated with Fusarium culmorum. Biochar at 10 t ha-1 increased grain yield by 23% relative to the control and significantly reduced crown rot severity. These effects were attributed to improved soil water-holding capacity. We recommend that farmers apply 10 t ha-1 biochar to suppress crown rot and raise yields.

1. Introduction

Fusarium crown rot causes yield losses in wheat in many dry production regions, and options for its management are limited. Biochar has been reported to change soil physical properties and microbial communities (Smith et al., 2019; synthetic placeholder), but evidence for effects on crown rot in wheat is scarce. Our objective was to quantify the effect of biochar rate on grain yield and crown rot severity under controlled conditions.

2. Materials and methods

2.1 Plant material and growth conditions

Winter wheat (Triticum aestivum L.) cultivar 'Alpha' (certified seed, lot 2023-117) was sown in 5 L pots filled with a sandy loam (pH 6.8; organic carbon 11.2 g kg-1, determined by dry combustion). Plants were grown in a greenhouse at 18/12 degrees C (day/night) with natural light supplemented to a 14 h photoperiod.

2.2 Treatments

Four biochar rates (0, 5, 10 and 20 t ha-1 equivalent) were mixed into the soil. Each biochar rate was applied to one greenhouse bench, and six pots per rate were placed on that bench. Pots were not rearranged during the experiment.

2.3 Inoculation and disease assessment

Plants were inoculated with Fusarium culmorum at tillering. Crown rot severity was scored on a 0-5 scale by one assessor at 14, 28, 42 and 56 days after inoculation. Grain yield per pot was determined at maturity and converted to t ha-1.

2.4 Statistical analysis

Data were analysed by one-way ANOVA with biochar rate as the factor, treating each pot-by-date observation as a replicate. Treatment means were compared using Fisher's LSD test at P < 0.05 for all pairwise comparisons. Analyses were done in R.

3. Results

3.1 Grain yield

Biochar rate affected grain yield (F(3,20) = 2.91, P = 0.012; Table 1). Yield at 10 t ha-1 (5.71 t ha-1) exceeded the control (4.82 t ha-1), an increase of 18.5%.

Biochar rate (t ha-1)	Grain yield (t ha-1)	LSD group
0 (control)	4.82 ± 0.31	a
5	5.12 ± 0.28	ab
10	5.71 ± 0.30	b
20	5.39 ± 0.33	ab

Table 1. Grain yield of wheat by biochar rate (n = 6 pots per rate).

3.2 Crown rot severity

Crown rot severity decreased with increasing biochar rate (F(3,92) = 6.84, P < 0.001; Fig. 1), with mean scores of 3.9, 3.4, 2.6 and 2.8 for 0, 5, 10 and 20 t ha-1, respectively.

Figure 1. Mean crown rot severity score by biochar rate (image not included in this synthetic example).

3.3 Soil pH

Soil pH at harvest did not differ among biochar rates (F(3,20) = 0.84, P = 0.49).

4. Discussion

The increase in yield and the reduction in crown rot are explained by improved soil water-holding capacity under biochar, which reduced drought stress and thereby root rot. Soil pH was not altered by biochar, so a liming effect is unlikely to explain the results.

Because only one cultivar and one isolate were used in a single greenhouse experiment, the results may not transfer to field conditions. These results show that biochar should be applied at 10 t ha-1 by farmers to control crown rot and increase yield.

5. Conclusions

Biochar at 10 t ha-1 improves yield and suppresses crown rot in wheat.

References

Smith, J., Lee, K., Rao, P. (2019). Biochar and soil water retention. [synthetic placeholder reference]